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Reflection Journal

Part 1: Brian; Introduction + Experience with Standard Scaling + Personal Learning

Module 05 focused on data preparation in machine learning and how to get data ready before training a model. Data preparation includes cleaning, organizing, and transforming data so a model can use it effectively. It matters because poorly prepared data can make it harder for the model to learn patterns and produce reliable results. In this lab, I focused on working with standard scaling. Standard scaling adjusts the numerical values so that the mean is close to zero, and the standard deviation is one. I worked with this technique to understand how numerical features such as Age and Fare can be transformed. This experience helped me understand why data preparation an important step in the machine is learning process.

When I worked with standard scaling, I learned that the changes in numerical values are based on their relationship with average and standard deviation. Age and Fare have different numerical ranges, so scaling can help put these features on a scale that is comparable to the other values. Running the code cell helped me see how the technique works instead of only reading its definition. Reviewing the output helped me understand how the numerical values changed after scaling. One thing I learned was that scaling values can be negative or positive depending on the original value. Values below or above average. This lab taught me that understanding the code

and its results is important because it helps me apply data preparation techniques correctly in future machine learning projects.

Part 2: Toan; Median imputation, missing data, and challenges + Personal Learning.

Part 2: Toan; Median Imputation, Missing Data, and Challenges + Personal Learning

In Module 05, I learned about missing values and how to handle them using median imputation. Missing values are data points that do not have a recorded value in the dataset. They can cause problems during data preparation because incomplete data may affect the accuracy of analysis and make it harder for some machine-learning algorithms to process the dataset. Median imputation is a technique that fills missing numerical values using the median, which is the middle value when the data is arranged in order. During the lab, I worked with the median-imputation code and reviewed the output to understand how missing values were handled. I learned that this technique can help keep the dataset complete without removing entire rows that contain missing information. Checking the output also helped me understand why we need to examine the dataset after applying a data-preparation technique to make sure the missing values were handled as expected.

One thing I learned from this lab was that understanding the purpose of each code cell is just as important as running the code. I needed to connect the code with the concepts we were learning and understand how the output showed the results of the data-preparation process. Working with my teammates helped me improve my understanding by allowing us to review the notebook and discuss the results together. I also realized that reviewing the output and adding

comments can make the code easier to understand when looking back at it later. If I complete a similar lab in the future, I would spend more time explaining what each step does and checking the dataset before and after applying the technique. Overall, this lab helped me understand why handling missing data is important and how median imputation can prepare a dataset for further analysis and machine-learning tasks.

Part 3: Montserrat; Categorical encoding, future applications + conclusion

In Module 04, I learned how to explore the Titanic dataset using Pandas, Matplotlib, and Seaborn. These tools helped me examine the data, identify missing values, and discover patterns. Module 05 builds on these skills by teaching me how to prepare data for machine learning through imputation, encoding, and scaling.

Categorical encoding converts these values into numerical representations that machine-learning algorithms can process. The encoded Data Frame represents the dataset after encoding. Reviewing the output helped me understand how categorical information is transformed. These techniques will help me prepare data for future machine-learning projects and understand the importance of data preparation in the machine-learning workflow.

Conclusion

Module 05 taught us important data-preparation techniques. Median imputation fills out missing numerical values; categorical encoding converts categories into numerical representations, and standard scaling puts numerical features on a comparable scale. Running all the code cells helped our team understand how these steps work together. Overall, the lab

prepared us for future machine-learning assignments by teaching us how to clean and transform data before training models.